Dakt 12/9/26

Exno:11

Routing at Network Layer

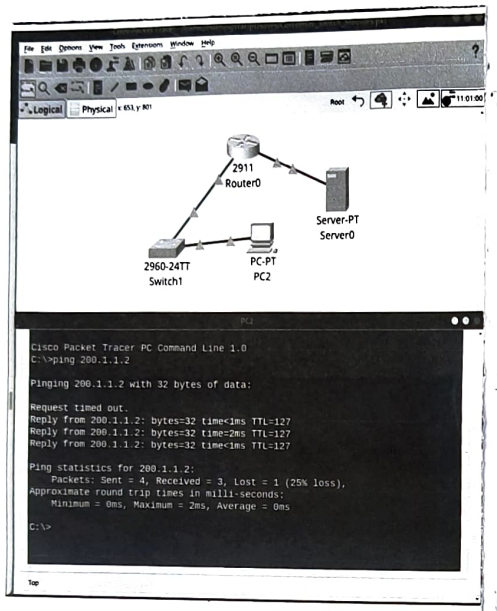
a) Simulate static Routing Protocol Configuration using Cisco Packet Tracer.

* Aim:

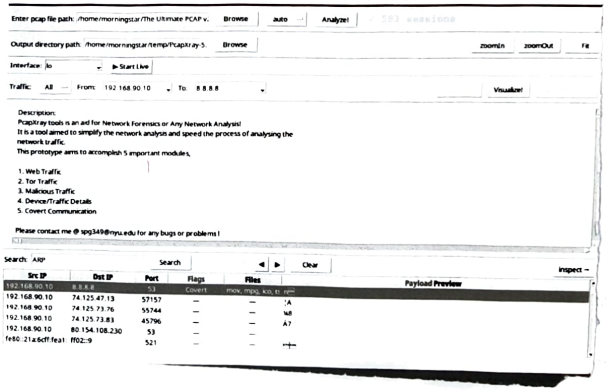
To simulate and configure static Routing Protocol using Cisco Packet Tracer and observe packet routing between different networks.

* Procedure:

1) Open Cisco packet Tracer.
2) place three routers, two switches and four PCs.
3) Connect the devices using appropriate Ethernet cabel
4) Enable the router interface using the no shutdown command.
5) Configure static routes on the routers using IP route command.
6) Configure a primary route and backup route using different administrative distance.
7) Verify using routing table show ip route static.
8) Test Connectivity between PCs using the ping command
9) Use Tracert <destination-IP> to observe destination
10) Simulate the failure the main route and verify that the backup route is used.



```
→ tshark -r "$F" -Y "dns.flags.response==0" -T fields -e ip.src -e dns.qry.name -e dn
s.qry.type | sort | uniq -c | sort -rn | head -20
     53 172.16.81.99    v10.events.data.microsoft.com   28
     44 172.16.81.99    v10.events.data.microsoft.com   1
     25         iTunes_Ctrl_73F532408528F239._dacp._tcp.local   255
     25 192.168.7.26    iTunes_Ctrl_73F532408528F239._dacp._tcp.local   255
     24 172.16.81.99    settings-win.data.microsoft.com 28
     21 172.16.81.99    wpad.ad2024.weberlab.de 1
     20 172.16.81.99    wpad.ad2024.weberlab.de 28
     18 172.16.81.99    settings-win.data.microsoft.com 1
     18 172.16.81.99    cp601.prod.do.dsp.mp.microsoft.com      28
     17 172.16.81.99    3.tlu.dl.delivery.mp.microsoft.com      28
     17 172.16.81.99    3.tlu.dl.delivery.mp.microsoft.com      1
     15 192.168.121.2   blog.webernetz.net      1
     15 192.168.110.9   blog.webernetz.net      1
     15 172.16.81.99    vm39-ad2024.ad2024.weberlab.de  1
     15 172.16.81.99    cp601.prod.do.dsp.mp.microsoft.com      1
     14         _homekit._tcp.local,_airplay._tcp.local,_sleep-proxy._udp.local12,12,
12
     14 85.215.94.29    router1.weberlab.de     1
     14 192.168.7.26    _homekit._tcp.local,_airplay._tcp.local,_sleep-proxy._udp.loc
al 12,12,12
     13 172.16.81.99    g.live.com      28
     12 172.16.81.99    g.live.com      1
```



* Result:

Thus, the Static Routing was successfully simulated and configured using cisco packet Tracer through primary and backup routes was verified.